

Customer Segmentation Analysis Report

Introduction

Customer segmentation is an important business strategy used to understand customer behavior and improve marketing effectiveness. This project applies K-Means Clustering to group customers based on purchasing patterns and demographic characteristics.

Objective

The primary objective of this project is to identify distinct customer segments and provide business recommendations for targeted marketing campaigns.

Data Preprocessing

The following preprocessing steps were performed:

- Checked missing values.
- Removed duplicate records.
- Dropped unnecessary columns.
- Detected and removed outliers.
- Standardized numerical features.

Exploratory Data Analysis

Several visualizations were created to understand customer behavior:

- Income Distribution
- Age Distribution
- Correlation Heatmap
- Spending Analysis
- Marital Status Analysis
- Relationship Status Analysis

Feature Engineering

Two new features were created:

Marital

A new feature representing customer marital status:

- Married
- Together
- Single
- Divorced

- Widow

In_relationship

A binary feature:

- 1 = Married or Together
- 0 = Single, Divorced, or Widow

K-Means Clustering

The following steps were used:

1. Standardization of data
2. Principal Component Analysis (PCA)
3. Elbow Method
4. Silhouette Score Analysis
5. Customer Segmentation

Optimal Number of Clusters

Based on Elbow Method and Silhouette Score Analysis, the optimal number of clusters was selected as:

K = 4

Cluster Characteristics

Cluster 0

- Medium to high income customers
- Higher spending behavior
- Significant number of customers in relationships

Cluster 1

- Lower income customers
- Lower purchasing activity
- Budget-conscious customer segment

Cluster 2

- Moderate income customers
- Average spending behavior
- Occasional purchasers

Cluster 3

- Highest income customers
- Highest spending behavior
- Premium customer segment

Key Findings

- Income strongly influences spending behavior.
- High-income customers contribute significantly to revenue.
- Relationship status affects purchasing patterns.
- Customer groups show clearly different spending habits.
- K-Means clustering successfully identified meaningful customer segments.

Recommendations

- Target premium customers with exclusive offers.
- Provide loyalty rewards to high-spending customers.
- Offer discounts and promotions to low-spending segments.
- Develop personalized marketing campaigns for each cluster.
- Focus retention efforts on valuable customer groups.

Conclusion

The customer segmentation analysis successfully identified four distinct customer groups. These insights can help businesses improve marketing strategies, increase customer engagement, and maximize revenue through targeted decision-making.